Sewing Machine



Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **High** 7 labs

Covers these lab items: Feed of arm machine; High speed sewing machine; Lock stitch machine; Sewing Machine; Sewing Machines with tables; Zigzag sewing machine

What it is used for

A sewing machine joins fabric pieces using stitches formed by a needle and thread. Correct threading, fabric control, and safe machine operation help produce consistent seams and prevent needle injuries.

Main parts

- Needle carries the upper thread through the fabric.
- Presser foot holds the fabric against the feed mechanism.
- Feed dogs move the fabric beneath the presser foot.
- Needle plate supports the fabric around the needle opening.
- Thread spool and guides direct thread toward the needle.
- Bobbin and bobbin case supply the lower thread on applicable machines.
- Handwheel and speed control allow controlled needle movement and sewing speed.

Before you start

1. Read the machine instructions and identify the controls before starting.
2. Place the machine on a stable table with adequate lighting and workspace.
3. Inspect the needle, presser foot, thread path, bobbin, plug, and power cord for damage.
4. Switch off the machine before threading, changing the needle, or adjusting the presser foot.
5. Select a needle, thread, and stitch suitable for the fabric and task.
6. Keep fingers, loose clothing, and long hair away from the needle area.

How to use

1. Raise the needle and presser foot using the designated controls or handwheel.
2. Thread the upper thread and bobbin according to the machine's threading diagram.
3. Place the fabric beneath the presser foot and lower the foot onto the fabric.
4. Turn the handwheel slowly to confirm the needle moves freely before powered sewing.
5. Connect power and select a suitable stitch and speed for the material.
6. Hold the thread ends briefly and guide the fabric without pulling it.
7. Use the foot pedal or speed control to sew at a steady pace.
8. Stop with the needle in a safe position and raise the presser foot before removing fabric.
9. Cut the threads using the designated cutter or suitable scissors.
10. Switch off and disconnect power before cleaning lint or carrying out adjustments.

⚠ Safety

- Keep fingers away from the moving needle and presser foot during sewing.
- Never sew over pins or use a bent or damaged needle.
- Switch off power before threading, changing needles, or clearing tangled thread.
- Do not operate the machine with damaged wiring or missing safety guards.
- Use the handwheel only in the direction recommended by the manufacturer.
- Keep the machine and floor around the pedal dry and free from clutter.

Care & storage

- Remove lint from the bobbin area and feed dogs after switching off and disconnecting power.
- Replace blunt, bent, or damaged needles with the correct type.
- Follow the manual for lubrication, cleaning, and servicing requirements.
- Cover the machine and store accessories neatly in a dry place.

Common mistakes to avoid

- Threading the machine while it remains powered on.
- Pulling fabric forcefully while the needle is moving.
- Using the wrong needle or incorrect thread path for the fabric.
- Leaving lint or tangled thread in the bobbin area.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Prepared by - Subodh Kumar